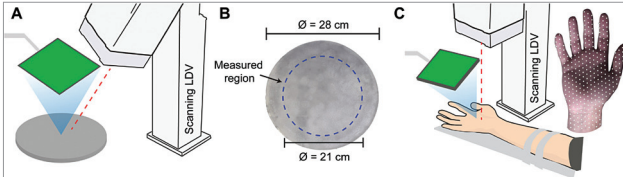




New holographic haptic displays can create better VR experiences

A team of researchers from UC Santa Barbara has discovered a new phenomenon that underlies emerging holographic haptic displays, which could lead to the creation of more compelling virtual reality experiences. Researchers used high-resolution optical imaging, simulations,

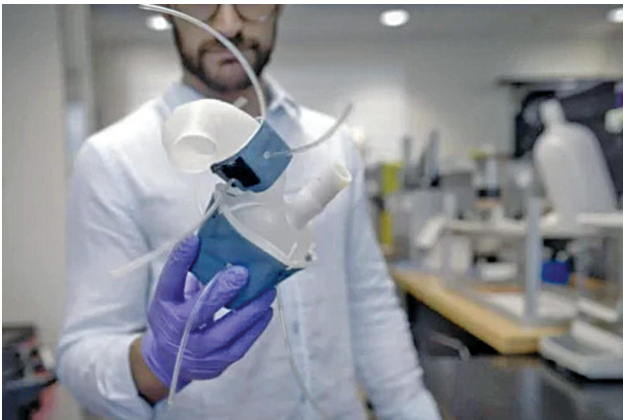


Time-resolved optical vibrometry characterisations were obtained during focused ultrasound stimulation of a tissue phantom and human hands (Credit: Science Advances, 2023)

and perception experiments to study ultrasound-excited waves that are excited in the skin during haptic holography. They discovered that holographic displays excite widespread vibration patterns—shear shock waves—in the skin. Shock waves create a trailing wake pattern that extends beyond the intended focal point, reducing the spatial precision and clarity of the tactile sensations. If the focused sound beam is a fast-moving boat on the water, the shock wave pattern is a wake trailing the boat. Current holographic haptic displays excite shock wave patterns that are so spread out in the skin that the sensations feel very diffuse.

A soft and flexible replica of a heart can now be 3D printed

Massachusetts Institute of Technology (MIT) engineers have developed a procedure to 3D print a soft and flexible

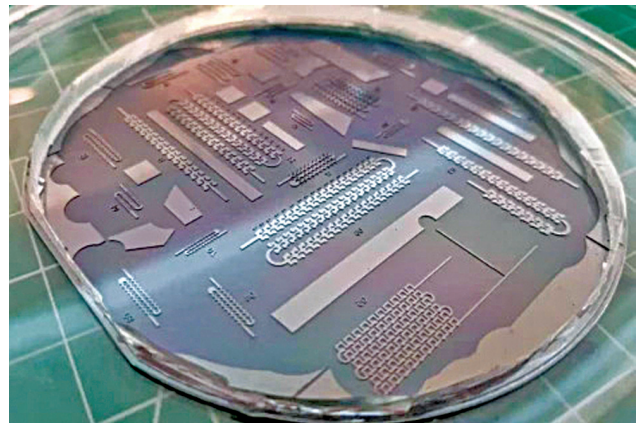


MIT engineers are hoping to help doctors tailor treatments to patients' specific heart form and function, with a custom robotic heart. The team has developed a procedure to 3D print a soft and flexible replica of a patient's heart (Credit: Melanie Gonick, MIT)

replica of a patient's heart. They can then control the replica's action to mimic that patient's blood-pumping ability. Researchers took advantage of 3D printing to produce custom replicas of actual patients' hearts. They used a polymer based ink that, once printed and cured, can squeeze and stretch, similarly to a real beating heart. They used medical scans of 15 patients diagnosed with aortic stenosis. The team converted each patient's images into a three-dimensional computer model of the patient's left ventricle and aorta. They fed this model into a 3D printer to generate a soft, anatomically accurate shell of both the ventricle and vessel. The team used an actuated printed heart to compare implants of different sizes, to see which would result in the best fit and flow—something they envision clinicians could potentially do for their patients in the future.

New device can detect and analyse cancer cells from blood samples

Researchers from the University of Technology Sydney (UTS) have developed a new device that can detect and analyze cancer cells from blood samples, enabling doctors to avoid invasive biopsy surgeries and monitor treatment progress. The new detection technology has 38,400



The Static Droplet Microfluidic device is able to rapidly detect circulating tumor cells that have broken away from a primary tumor and entered the bloodstream (Credit: Dr Majid Warkiani)

chambers capable of isolating and classifying the number of metabolically active tumor cells. Once the tumor cells are identified with the device, they can undergo genetic and molecular analysis, which can aid in the diagnosis and classification of the cancer and inform personalised treatment plans. This new technology is designed for integration into research and clinical labs without relying on high-end equipment and trained operators. This will enable doctors to diagnose and monitor cancer patients in a practical and cost-effective manner.